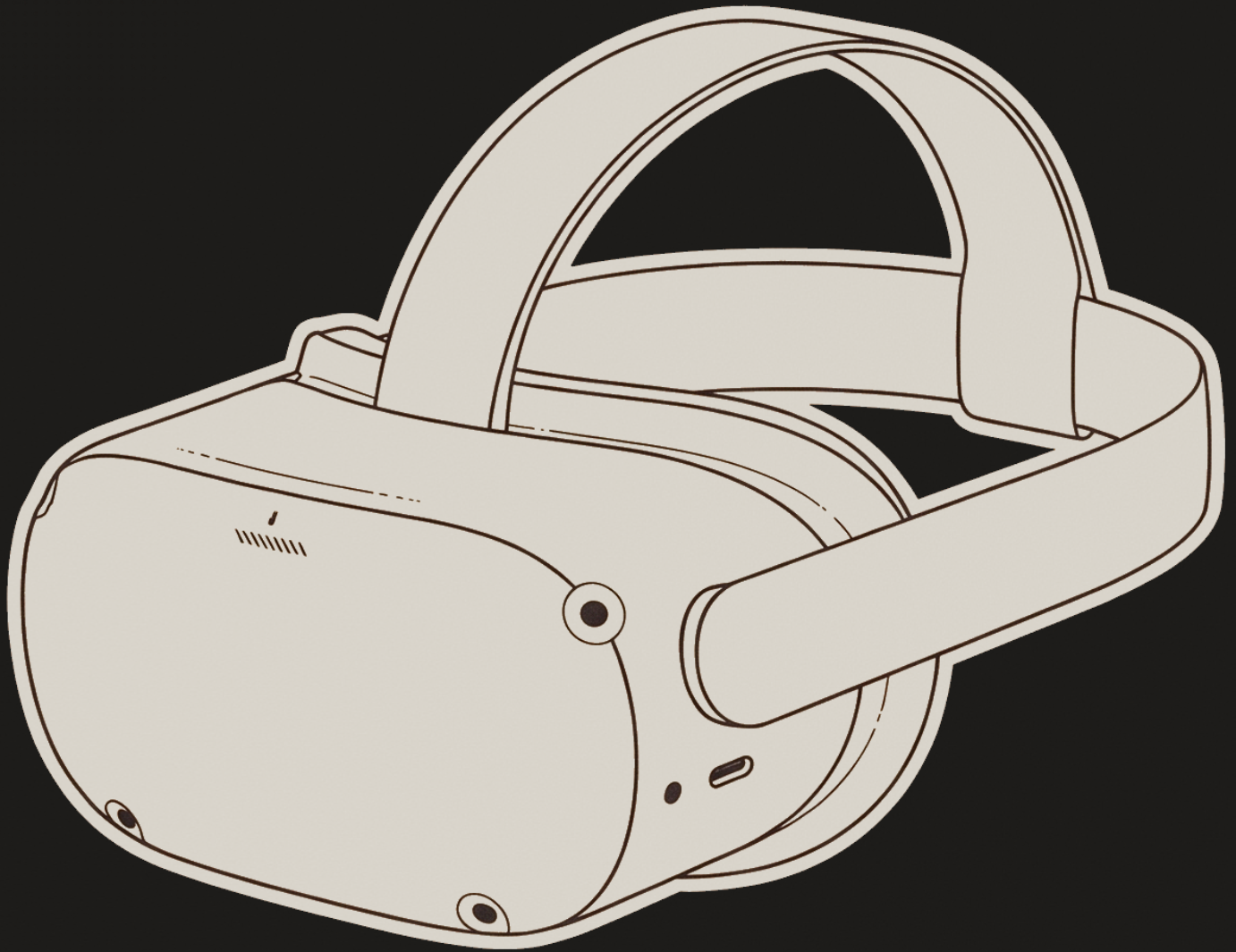


OMANG BAHETI

SOFTWARE/XR DEVELOPER | UX RESEARCHER | GAME DESIGNER



Building immersive experiences in virtual, augmented & mixed reality –
researched, designed, and shipped for the arcade of tomorrow.

[VIEW WORK](#)

[CONTACT ME](#)

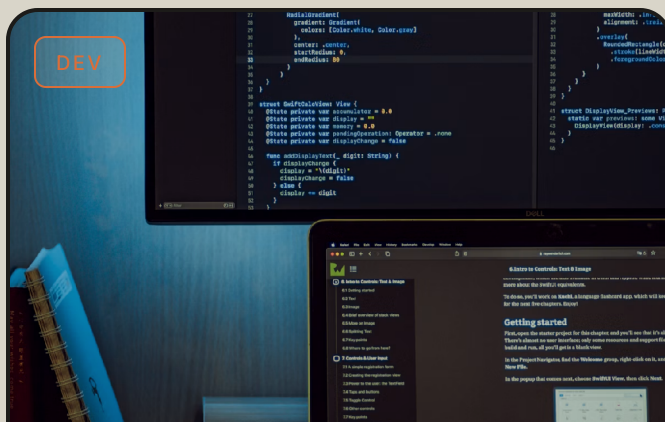


FEATURED WORKS

F1 ALL

F2 DEV

F3 RESEARCH



001

2024

VOID PROTOCOL

VR PSYCHOLOGICAL HORROR

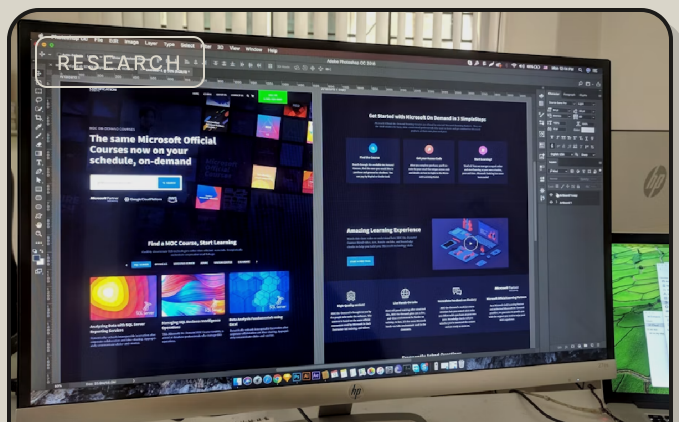
Single-player horror, 6DOF. Three months of presence research with 42 participants shaped every design decision before an asset was built.

VR

6DOF

SPATIAL AUDIO

HAPTICS



002

2024

XR ONBOARDING PATTERNS

MULTI-PLATFORM RESEARCH PROGRAMME

12 weeks, 78 participants, 5 platforms. Public pattern library now adopted by three studios. Presented at CHI 2024.

USER STUDIES

PATTERN LIBRARY

SPATIAL UX



DEV

003

2024

NEXUS ARENA

5V5 COMPETITIVE MULTIPLAYER

Custom rollback netcode. Four embedded playtest cohorts drove a full UI redesign between alpha and ship.

MULTIPLAYER

ROLLBACK NETCODE

PLAYTESTING



RESEARCH

004

2023

SPATIAL MENTAL MODELS

EYE-TRACKING STUDY — PASSTHROUGH AR

60 participants. Heatmaps, task-completion analysis, and retrospective interviews across two headset generations.

EYE TRACKING

THINK-ALOUD

PASSTHROUGH AR



DEV

005

2023

TERRAIN AR

GEOSPATIAL FIELD RESEARCH TOOL

LiDAR-powered geospatial overlay for field researchers. Validated through contextual inquiry with 18 domain experts.

AR

LIDAR

GEOSPATIAL

RESEARCH LOG

PUBLICATIONS



REDUCING COGNITIVE LOAD IN ROOM-SCALE XR WORKSPACES

F. LASTNAME, A. COAUTHOR, B. COAUTHOR

CHI '25 — ACM Conference on Human Factors in Computing Systems

XR

HCI

STUDY

DOI ↗

DETAILS ↗



DWELL-TIME SIGNATURES AS AN INDICATOR OF INTERFACE AMBIGUITY

F. LASTNAME, C. COAUTHOR

ISS '24 — ACM Interactive Surfaces and Spaces

UX RESEARCH

EYE TRACKING

DOI ↗

DETAILS ↗



LOCAL-FIRST ARCHITECTURES FOR OFFLINE FIELD RESEARCH TOOLS

F. LASTNAME

VL/HCC '24 — IEEE Symposium on Visual Languages and Human-Centric Computing

SOFTWARE ENGINEERING

DETAILS ↗



AMSTERDAM, NL

2024

PLAYER PROFILE

ABOUT

I'm Omang Baheti — XR developer and UX researcher, eight years building and studying immersive technology from both sides of the monitor.

Research gives engineering decisions an empirical anchor. Engineering makes research actionable — a usability finding can become a pull request the same day.

Previously at [Niantic Labs](#), [Ready at Dawn](#), and [Microsoft Research](#).
Now independent.

8+

YEARS XR

200+

PARTICIPANTS

22

SHIPPED

3

GDC TALKS

XR DEV

- ▶ Unreal Engine 5
- ▶ Unity 6
- ▶ OpenXR / WebXR
- ▶ HLSL / Shader Graph
- ▶ Rollback Netcode

UX RESEARCH

- ▶ User Interviews
- ▶ Usability Testing
- ▶ Think-Aloud
- ▶ Eye Tracking
- ▶ Diary Studies

ANALYSIS

- ▶ Thematic Coding
- ▶ Affinity Mapping
- ▶ Journey Mapping
- ▶ Statistical Analysis
- ▶ Heuristic Eval

TOOLS

- ▶ Figma / FigJam
- ▶ Dovetail
- ▶ C++ / C# / Python
- ▶ SPSS / R
- ▶ Miro

HOW I WORK

PROCESS

■ DEVELOPER

I build in Unreal and Unity – shaders, netcode, spatial interaction, performance profiling. I own the implementation, not just the spec.

■ RESEARCHER

I plan and run user studies, analyse qualitative and quantitative data, and translate findings into decisions that actually ship.

01

RESEARCH FIRST

Every project begins with understanding people – who they are, what they expect, and where current systems fail them.

02

PROTOTYPE IN ENGINE

Paper wireframes don't capture spatial feel. I prototype in Unity or Unreal early so decisions are grounded in real physics and latency.

03

TEST IN CONTEXT

Lab studies miss the messy middle. Sessions run in living rooms, offices, and public spaces.

04

SHIP & LOOP

Post-launch telemetry and follow-up studies feed the next cycle. Publishable findings go back into the community.

START A FIGHT

START A PROJECT

Open for XR development, embedded research, and hybrid engagements where both disciplines matter.

EMAIL	OMANG@BAHETI.DEV
LOCATION	AMSTERDAM / REMOTE
DEV SLOT	Q1 2025 – OPEN
RESEARCH	Q1 2025 – LIMITED
RESPONSE	< 24 HOURS

NAME

EMAIL

Your name

your@email.com

ENGAGEMENT TYPE

VR EXPERIENCE

AR APPLICATION

GAME DEV

UX RESEARCH

EMBEDDED RESEARCH

MIXED REALITY

CONSULTING

MESSAGE

Platform, research goals, timeline...

SEND FIGHT →

